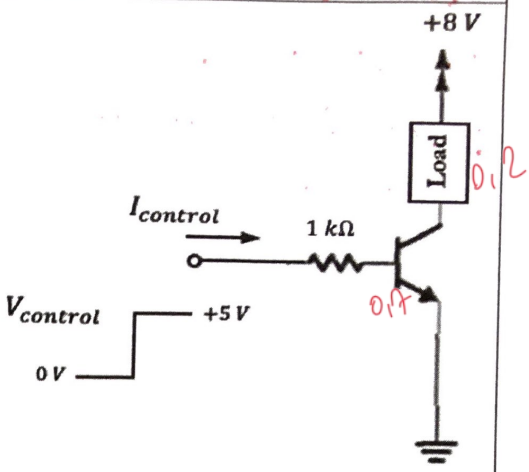
Name: KEY	ID:	Mark: /100

1) (30 points) In the given circuit, the BJT has a base-emitter ON voltage of 0.7 V, a base-collector ON voltage of 0.5 V and β=50. Assume that the load is a lamp. The circuit is supposed to switch the lamp ON or OFF based on a digital control signal ($V_{control}$, see figure). Answer the following questions with full details.



a) The maximum possible voltage across the lamp is (circle one)
(a) +8 V (b) +7.8 V (c) +7.5 V (d) +5 V (e) +4.3 V.

Explain:

It happens when transistor is saturated.

$V_C = 0.2\,V \Rightarrow V_{load_{max}} = 8 - 0.2 = 7.8\,V$

b) What are the values of $V_{control}$ and $I_{control}$ to achieve the voltage in part (a)?

Maximum load voltage can only happen when transistor saturated

$V_{control} = 5\,V \Rightarrow I_{control} = \frac{5-0.7}{1K} = 4.3\,mA$

c) Show that the minimum lamp resistance to achieve the voltage in part (a) is about 36 Ω.

$I_B = I_{control} = 4.3mA \Rightarrow I_C = 50 \times 4.3m = 0.215\,A$ } Edge of saturation – active mode

$R = 7.8/0.215 \approx 36\,\Omega$

d) If the lamp resistance is 50 Ω, what is the maximum power consumed by the lamp?

$R_{lamp} > 36\Omega \Rightarrow$ saturation $\Rightarrow P = V^2/R = (7.8)^2/50 = 1.22\,W$

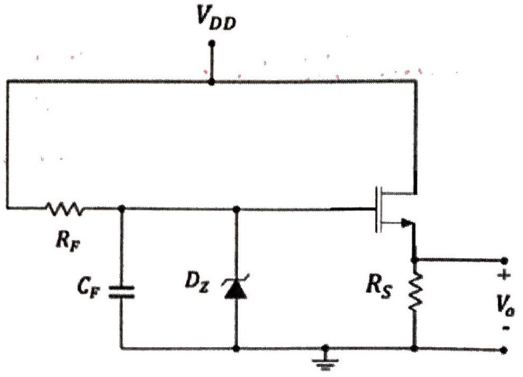
e) If the lamp resistance is 15 Ω, what is the maximum power consumed by the lamp?

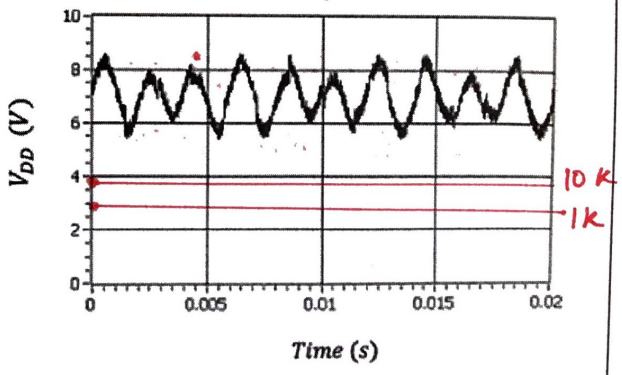
$R_{lamp} < 36\Omega \Rightarrow$ active mode $\Rightarrow I_C = 0.215\,A \Rightarrow P = (0.215)^2(15)$

$= 0.69\,W$

2) (30 points) The DC biasing voltage V_{DD} applied to the given circuit has the waveform shown in the figure. The Zener diode has V_{zo} = 5.3 V, $r_z = 0\,\Omega$ and I_{z_min} =.5 mA. The NMOS has $V_t = 1$ V, k=1 mA/V², and λ=0. Take $R_F = 1\,k\Omega$ and $C_F = 100\,\mu F$. Give full details of the solution.

a) On the same graph, sketch the waveform of the output V_o for $R_S = 1\,k\Omega$. Show all labels.
b) Repeat for $R_S = 10\,k\Omega$.





Zener always ON,

So $V_G = 5.3\,V$

$V_D = 5.5 - 8.5\,V$

Check saturation

V_D ? $V_G - V_t$

$5.5 > 5.3 - 1$

$\Rightarrow$ transistor always saturated

Current $I_D = K(V_{GS} - V_t)^2 = \frac{1}{2}m(5.3 - V_o - 1)^2$

$V_o = R_s I_D \Rightarrow V_o = \frac{1}{2}(m) R_s (4.3 - V_o)^2$

For $R_s = 1K\Omega \Rightarrow V_o^2 - 9.6\,V_o + 4.3^2 = 0$

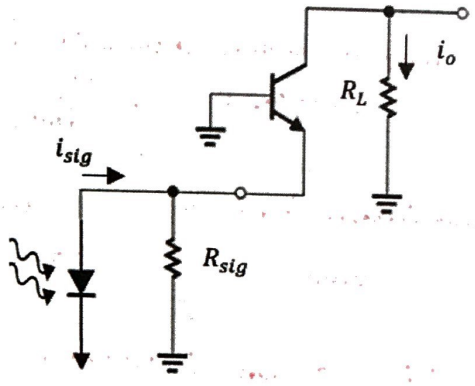
$V_o = 2.7\,V$

For $R_s = 10K\Omega \Rightarrow V_o^2 - 8.7\,V_o + 4.3^2 = 0$

$V_o = 3.7\,V$

3) (40 points) The input to the amplifier circuit below is a weak photodiode current i_{sig} (considered as an independent ac current source). The photodiode has an internal resistance R_{sig}, as shown. Assume that the transistor is in active mode, ignore r_o and,

a) Draw the amplifier equivalent circuit using the T-model for the BJT.
b) Show that the amplifier current gain $A_i = \alpha$.
c) Find the overall amplifier current gain.
d) Find the amplifier input resistance.
e) Find the amplifier output resistance.
f) Show that the amplifier voltage gain $A_v = g_m R_L$.
g) Based on your results, does this circuit serve as a "good" current amplifier for this application? Explain.



b) $i_o = -\alpha i_e$, $i_i = -i_e \Rightarrow \frac{i_o}{i_i} = \alpha$

c) $\frac{i_o}{i_{sig}} = \frac{i_o}{i_i} \cdot \frac{i_i}{i_{sig}} = \alpha \frac{R_{sig}}{R_{sig} + R_i}$

d) $R_i = re$

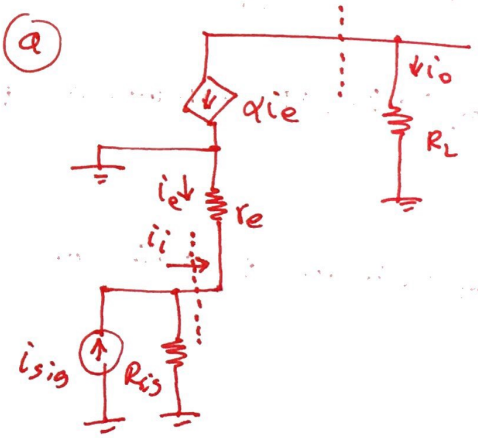
e) $R_o = \infty$

f) $\frac{v_o}{v_i} = \frac{i_o}{i_i} \frac{R_L}{R_i} = \alpha \frac{R_L}{re}$

e) R_i is low (Good) but gain $A_i \leq 1$ (not suitable for this application)

R_o is high (Good)

(a)



MOSFET Current Equations	Triode	$I_D = k\,[2(V_{GS} - V_t)V_{DS} - V_{DS}^2]$	Saturation	$I_D = k\,(V_{GS} - V_t)^2(1 + \lambda V_{DS})$
BJT Current Equations	Active	$I_C = \beta I_B, I_E = (\beta + 1)I_B, I_C = \alpha I_E$	Relations	$\alpha = \beta/(\beta + 1)$
Diode Current Equations	Forward	$I_D = I_s\,(e^{V_D/nV_T} - 1)$	Roots	$x = (-b \pm \sqrt{b^2 - 4ac})/2a$
Amplifier Equations	Gain	$A_v = v_o/v_i,\quad A_i = i_o/i_i$	Resistance	$R_i = v_i/i_i,\quad R_o = v_o/i_o$